Class-restricted ℓ^2-congruences for overpartitions modulo 13, 17 and 23

Anonymous

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Abstract

We prove several new families of Ramanujan-type congruences for the overpartition function $\overline{p}(n)$, for example

$$\overline{p}(17 \cdot 4^i \cdot 29^2 \cdot m) \equiv 0 \pmod{17} \quad \text{for all } i \geq 0,\ m \equiv 3 \pmod 8,\ \left(\frac{m}{29}\right) = -1.$$

The displayed ℓ^2-families of Ryan, Sirolli, Villegas-Morales and Zheng for these moduli have Legendre restrictions but no additional class restriction modulo 8. Here small primes ℓ occur with such a class restriction. A filtered finite search suggested the examples. The proofs use Shimura's operators on low-dimensional modular-form spaces and correction terms that are reductions of exact $U(4)$ eigenvectors. One support identity therefore proves every tower level. We also classify exactly the levels at which the whole class $13 \cdot 4^i(8n+7)$ vanishes modulo 13.

1 Introduction

Let $\overline{p}(n)$ count the overpartitions of n [2], so that $\sum \overline{p}(n)q^n = 1/\varphi(-q)$ with $\varphi(q) = \sum_{k\in\mathbb{Z}} q^{k^2}$. Treneer [8] showed that $\overline{p}$ satisfies congruences $\overline{p}(m\ell^3 n) \equiv 0 \pmod m$, and Ryan, Sirolli, Villegas-Morales and Zheng [5] (also Zheng [10]) made these explicit. RSVZ give families of two kinds: $\overline{p}(m\ell^3 n) \equiv 0 \pmod m$, and $\overline{p}(m\ell^2 n) \equiv 0 \pmod m$ for n prime to ℓ with a single Legendre condition. Both kinds are infinite in ℓ for $m \in \{3,5,7,11\}$. For $m \in \{13,17,19\}$ they list the primes ℓ found by a search up to about 5000; in the ℓ^2-families the smallest are $\ell = 431$ for $m = 13$ and $\ell = 167$ for $m = 17$. For $m = 23$ they remark on $\ell = 5303$ and 8783. They note that cusp forms make the case $m \geq 13$ harder. None of these families has a congruence condition modulo 8. Wei [9] proved $\overline{p}(11(8n+5)) \equiv 0 \pmod{11}$ and $\overline{p}(13 \cdot 2^6(8n+7)) \equiv 0 \pmod{13}$. He also conjectured congruences modulo 7, 17, 19 and 23 involving residue classes modulo 8 together with Legendre conditions. Those modulo 17 and 23 are the cases $i = 0$ of (I) (with $\ell = 11$) and of (II) below, and they are proved here. A different treatment, together with the modulo-19 case, is in the modulo-17/19/23 paper in this versioned collection.

The discovery scan used $p \in \{13,17,19,23,29,31\}$, $\ell \in \{3,5,7,11,13,17,19,23,29,31,37,41,43\}$ with $\ell \neq p$, $0 \leq j \leq 6$, odd classes modulo 8, and both nonzero Legendre signs. Arguments were strictly below 40000000. It retained a candidate only after at least forty matching instances and discarded classes already vanishing without the ℓ^2 factor. Thus it was a filtered search, not an exhaustive classification of all possible congruences even within the parameter box. The logs are historical discovery evidence; the proofs below do not extrapolate them.

Theorem 1. *For all integers $i \geq 0$ and $m \geq 1$ satisfying the conditions below:*

(I) $\overline{p}(17 \cdot 4^i \ell^2 m) \equiv 0 \pmod{17}$ *for* $\ell \in \{11, 29\}$, $m \equiv 3 \pmod 8$, $\left(\frac{m}{\ell}\right) = -1$;

(II) $\overline{p}(23 \cdot 4^i \cdot 13^2 m) \equiv 0 \pmod{23}$ *for* $m \equiv 5 \pmod 8$, $\left(\frac{m}{13}\right) = 1$;

(III) $\overline{p}(13 \cdot 4^i \cdot 7^2 m) \equiv 0 \pmod{13}$ *for* $m \equiv 7 \pmod 8$, $\left(\frac{m}{7}\right) = -1$.

Moreover:

(IV) $\overline{p}(52\,\ell^2 m) \equiv 0 \pmod{13}$ *for* $\ell \in \{5, 29\}$, $m \equiv 1 \pmod 4$, $\left(\frac{m}{\ell}\right) = -1$; *and* $\overline{p}(104 \cdot 5^2 m) \equiv 0 \pmod{13}$ *for all odd* m *with* $\left(\frac{m}{5}\right) = 1$.

The cases $i = 0$ of (I) with $\ell = 11$ and of (II) are Wei's conjectures and coincide with the modulo-17/23 results in the companion paper in this collection; they are not separate contributions. The case $i = 3$ of (III) follows from Wei's theorem since $49m \equiv m \pmod 8$. In fact Proposition 4 below shows that precisely $i \equiv 3 \pmod 4$ have this whole-class explanation. The consulted literature did not establish the remaining exact parameter families; this bounded search is not universal priority evidence. The additional class conditions and actual prime choices, rather than square-prime shape alone, distinguish the displayed families.

2 Tools

Let $F = \sum_{n \text{ odd}} \sigma(n) q^n = \eta(4z)^8/\eta(2z)^4 \in M_2(\Gamma_0(4))$. For $\lambda \geq 0$ the space $V_\lambda = M_{\lambda+1/2}(\Gamma_0(4))$ has the basis $B_a = F^a \varphi^{2\lambda+1-4a}$, $0 \leq a \leq \lfloor (2\lambda+1)/4 \rfloor$, with $B_a = q^a + O(q^{a+1})$ [4, Thm. 1.49]. We use:

1. Shimura's operator [6] $T(\ell^2)$ (ℓ an odd prime) preserves V_λ, and acts by
$$(f \mid T(\ell^2))(n) = a(\ell^2 n) + \left(\frac{(-1)^\lambda n}{\ell}\right) \ell^{\lambda-1} a(n) + \ell^{2\lambda-1} a(n/\ell^2).$$

2. $U(4)$ preserves V_λ. Indeed, by [4, Prop. 3.7], $U(2)$ maps $M_{\lambda+1/2}(\Gamma_0(8), \psi)$ to $M_{\lambda+1/2}(\Gamma_0(8), \psi\left(\frac{8}{\cdot}\right))$, so $U(4) = U(2)^2$ preserves $M_{\lambda+1/2}(\Gamma_0(8)) \supseteq V_\lambda$. For each basis element, the difference D between $B_a \mid U(4)$ and the element of V_λ agreeing with it in the first $\dim V_\lambda$ coefficients satisfies $D\varphi \in M_{\lambda+1}(\Gamma_0(8), \chi_{-4}^{\lambda+1})$. This space has Sturm bound $\lambda+1$, and we checked $D\varphi = 0$ well beyond it. (This is also a theorem of Kohnen [3].)

3. The projection of $f \in V_\lambda$ onto an *odd* class modulo 8 lies in $M_{\lambda+1/2}(\Gamma_0(256))$, by averaging the four real characters [9, Prop. 2.4 and Remark 2.5]. Multiplying by φ gives weight $\lambda+1$ and character $\chi_{-4}^{\lambda+1}$, with Sturm bound $32(\lambda+1)$. The unit φ may then be cancelled. This character averaging does not project onto an even class.

4. (Wei's reduction [9].) Since $\varphi^p \equiv \varphi(q^p) \pmod p$, we have $\varphi \cdot \sum \overline{p}(pn)(-q)^n \equiv \varphi^{p-1} \mid T(p) \pmod p$, where the right side lies in $M_{(p-1)/2}(\Gamma_0(4), \chi_{-4}^{(p-1)/2})$. Let c_p be the element of $V_{(p-3)/2}$ with $c_p\varphi \equiv \varphi^{p-1} \mid T(p) \pmod p$; this is verified up to the Sturm bound on $\Gamma_0(4)$. Since φ is a unit in $\mathbb{F}_p[[q]]$, we have $\overline{p}(pN) \equiv (-1)^N c_p(N) \pmod p$. In the basis (B_a), c_p has coordinates modulo p equal to $(1, 4, 1)$ for $p = 13$, $(1, 13, 13, 0)$ for $p = 17$ and $(1, 9, 5, 14, 17, 20)$ for $p = 23$. These were verified up to the relevant Sturm bounds on $\Gamma_0(4)$ and directly against $\overline{p}$.

3 The mechanism

Fix p, $\lambda = (p-3)/2$ and $c = c_p$. Let $\mathcal{N}$ be a set of positive integers prime to ℓ on which $\eta := \left(\frac{(-1)^\lambda N}{\ell}\right)$ is constant. Put $h := c \mid T(\ell^2) - \eta\,\ell^{\lambda-1}c$. Comparing coefficients gives

$$c(\ell^2 N) \equiv h(N) \pmod{p} \qquad (N \in \mathcal{N}). \tag{1}$$

So it suffices to show that h vanishes on $\mathcal{N}$. For $\mathcal{N} = \{4^i m : \ m \equiv r\ (8)\}$ we use $h(4^i m) = (h \mid U(4)^i)(m)$. The coordinates of $h \mid U(4)^i$ modulo p run through a *finite* orbit under the $U(4)$-matrix. It therefore suffices to check, for each orbit element, that its projection to the class r modulo 8 vanishes, which is a single Sturm check on $\Gamma_0(256)$.

Lemma 2 (Support propagation)**.** *Suppose $c \mid T(\ell^2) - \epsilon\ell^{\lambda-1}c \equiv sH \pmod{p}$, $H \mid U(4) = \alpha H$, and $H(m) = 0$ on an odd class $r \pmod 8$. Then $c(\ell^2 4^i m) \equiv 0 \pmod{p}$ for every $i \geq 0$ and $m \geq 1$ in that class with $\left(\frac{(-1)^\lambda m}{\ell}\right) = \epsilon \in \{1, -1\}$.*

Proof. The Legendre condition ensures $\ell \nmid m$ and is unchanged by multiplication by 4^i. The backward Hecke term is therefore zero and the middle term cancels. The remaining coefficient is $sH(4^i m) = s\alpha^i H(m) = 0$. □

This elementary reusable lemma packages the existing Hecke argument; it is not claimed as a new general theorem in modular-form theory.

4 Proof of Theorem 1

The correction terms below are reductions of the following exact forms, written in the basis B_a of their respective spaces:

p	λ	coordinates of H	α	$h \bmod p$	zero class
13	5	$(0, 1, -16)$	16	$12H$	7
17	7	$(0, 1, -16, 0)$	-64	$12H$	3
23	10	$(0, 8, -383, 4032, 1280, -8192)$	-512	$2H$	5

Here $H \mid U(4) = \alpha H$ over the integers. Identify $U(4)$ in the integral triangular basis, then check the displayed eigenvector identities exactly. Projection of H to the listed odd class, multiplied by φ, has all coefficients zero through bounds $192, 256, 352$, respectively. Sturm over characteristic zero proves the exact support identities (or apply it modulo every prime). Hence Lemma 2 proves (I)–(III). The former orbit periods are simply the orders of $16, -64, -512$ modulo the corresponding primes, namely $3, 4, 22$.

The first form is $\eta(z)^2\eta(2z)^7\eta(4z)^2$, already listed by Aydin and Inam [1, Table 1, no. 43]. The second is $\eta(2z)^{27}/(\eta(z)^6\eta(4z)^6)$; negative eta exponents here do not imply cusp poles. These identities are auxiliary descriptions, not claims of novel eigenforms. The explicit modular checks below also provide an alternative finite-orbit proof.

(I) and (II). The relevant data modulo p are as follows; full matrices are in the evidence bundle.

p	ℓ	η	h (coordinates)	$U(4)$-orbit of h
17	11	$+1$	$(0, 12, 12, 0)$	period 4: $(0, 12, 12, 0) \to (0, 14, 14, 0) \to (0, 5, 5, 0) \to (0, 3, 3, 0)$
17	29	-1	$(0, 12, 12, 0)$	the same orbit
23	13	$+1$	$(0, 16, 16, 14, 7, 15)$	period 22

For $p = 17$ the sign is $\eta = \left(\frac{-1}{\ell}\right)\left(\frac{m}{\ell}\right)$. Since $\left(\frac{-1}{11}\right) = -1$ and $\left(\frac{-1}{29}\right) = 1$, we get $\eta = +1$ and -1 respectively; that is, $c \mid T(29^2) \equiv -29^6 c + h$. For $p = 23$, $\lambda = 10$ is even and $\eta = \left(\frac{m}{13}\right) = 1$. For $N = 4^i m$ the character value does not depend on i, since $\left(\frac{4^i}{\ell}\right) = 1$. The orbit is purely periodic and contains h itself, so it covers every $i \geq 0$. For every orbit element, the projection to the class $r = 3$ (respectively 5) modulo 8, multiplied by φ, vanishes modulo p for $n < 4000$. This exceeds the Sturm bounds 256 and 352 on $\Gamma_0(256)$. By (1), $\overline{p}(p\ell^2 4^i m) \equiv \pm c(\ell^2 4^i m) \equiv \pm h(4^i m) \equiv 0$.

(III). Here $p = 13$, $\lambda = 5$, $\ell = 7$, and $\eta = \left(\frac{-4^i m}{7}\right) = \left(\frac{-1}{7}\right)\left(\frac{m}{7}\right) = 1$. Then $h = c \mid T(49) - 7^4 c$ has coordinates $(0, 12, 3)$. Its $U(4)$-orbit modulo 13 is the 3-cycle (containing h, so it covers all i) $(0, 12, 3) \to (0, 10, 9) \to (0, 4, 1)$. Each projection to the class 7 modulo 8, times φ, vanishes modulo 13 beyond the Sturm bound 192.

(IV). For $N = 4m$ with $m \equiv 1 \pmod 4$ and $\left(\frac{m}{\ell}\right) = -1$, $\ell \in \{5, 29\}$, we have $\eta = \left(\frac{-1}{\ell}\right)\left(\frac{m}{\ell}\right) = -1$. So $h_\ell = c \mid T(\ell^2) + \ell^4 c$ (each $T(\ell^2)$-image was identified in the 3-dimensional space V_5 from its first coefficients and checked well beyond), with coordinates $(7, 9, 12)$ for $\ell = 5$ and $(5, 12, 3)$ for $\ell = 29$. Then $h_\ell(4m) = (h_\ell \mid U(4))(m)$, and the projection of $h_\ell \mid U(4)$ to the classes $1, 5$ modulo 8 vanishes (Sturm bound 192). For $N = 8m$ with m odd and $\left(\frac{m}{5}\right) = 1$, $\eta = \left(\frac{-8m}{5}\right) = \left(\frac{-1}{5}\right)\left(\frac{2}{5}\right)\left(\frac{m}{5}\right) = -1$, so again $c(25N) \equiv h_5(N)$. With $k = h_5 \mid U(4)$ we need $k(2m) \equiv 0$ for odd m. The form $w = k \mid U(2) - (k \mid U(4)) \mid V(2)$ lies in $M_{11/2}(\Gamma_0(8), \left(\frac{8}{\cdot}\right))$, and its coefficients are $k(2m)$ for m odd and 0 for m even. Moreover $w\varphi$ vanishes modulo 13 beyond its Sturm bound 6. Since $\overline{p}(13N) \equiv \pm c(N)$, this proves (IV). □

Remark 3. The congruences need their specified conditions; the finite search is not a classification of all other classes or primes. The exact eigenvector identities explain propagation without testing every orbit state. Other primes can be tested by the same finite-space calculation, but adding a list of numerical cases alone would not strengthen the theorem.

Proposition 4 (Exact whole-class classification modulo 13)**.** *For every integer $i \geq 0$,*

$$\left[\overline{p}(13 \cdot 4^i(8n + 7)) \equiv 0 \pmod{13} \text{ for every } n \geq 0\right] \iff i \equiv 3 \pmod 4.$$

Proof. In the row-coordinate convention, the exact matrix and initial vector are

$$U = \begin{pmatrix} 1 & 5280 & 42240 \\ 0 & 336 & 2816 \\ 0 & 20 & 192 \end{pmatrix}, \qquad c_{13} = (1, 4, 1) \pmod{13}.$$

Its orbit modulo 13 has length 12 and returns to its initial vector. For each of its twelve vectors, projection onto class 7, followed by multiplication by φ, is tested through the inclusive Sturm bound 192. It vanishes precisely for states $3, 7, 11$. For every other state the coefficient at 7 is the nonzero witness

$$c_{13}(4^i 7) \equiv 7 + 4 \cdot 5^i \pmod{13}.$$

This formula is checked on all twelve states and is periodic, since 5 has order 4 modulo 13. The sign in $\overline{p}(13N) \equiv (-1)^N c_{13}(N)$ does not affect vanishing. Thus both directions hold for every i, not just the discovery range. □

The classification was suggested by the supplied reviewer audit and checked in the revised collection. Wei supplies the $i = 3$ instance; the extension's literature status is bounded by the recorded search.

5 Verification

All computations use exact or modular arithmetic (python-flint). They were carried out by two separate scripts (for $p \in \{17, 23\}$ and $p = 13$) and confirmed by a third, independent checker written by a separate reviewer. Each congruence was also checked directly from $\overline{p} \bmod p$: up to arguments $3 \cdot 10^6$ for (I) and (II), and up to $4 \cdot 10^7$ in the original search, with no exceptions.

Collection and verification

This paper is part of the revised collection, version 1.1.0-candidate, DOI `https://doi.org/10.5281/zenodo.23319157`. Companion papers and exact verification code are archived together. The statements are unrefereed computer-assisted proof candidates; no formal verification or unaffiliated reproduction is asserted.

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